GlycoTwin-IN

A hybrid Bergman digital twin for two-hour postprandial hyperglycaemia risk

Digital Twin Challenge 2026 — Prototype and code submission. Happiest Health. Individual entry.

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Team size: 1

Simulation proof of concept. Not a medical device. Not for clinical use.

Problem

India has a large and still rising burden of diabetes. ICMR-INDIAB-17 estimated a weighted prevalence of 11.4% among adults 20 years and older, corresponding to about 101 million people with diabetes and 136 million with prediabetes (Anjana et al., Lancet Diabetes & Endocrinology, 2023). Post-meal glucose excursions are a routine management problem, but clinic HbA1c does not show what a specific meal will do in the next two hours.

The prototype asks a narrow question: given a synthetic EHR snapshot and a wearable-like stream, will glucose reach 180 mg/dL within 120 minutes of a logged meal? The 180 mg/dL line is an operational flag aligned with a commonly cited 1–2 hour post-meal target, not a diagnostic claim.

Why a digital twin, not only a classifier

- The physical entity is an adult with type 2 diabetes. In this build that entity is simulated.
- The twin state is plasma glucose and remote insulin action, initialised from the EHR prior.
- The last 60 minutes of CGM update insulin sensitivity. A declared carbohydrate load is then rolled forward 120 minutes.
- The clinician sees a trajectory and a flag, not only a probability.
- A gradient-boosting classifier was compared. On the held-out synthetic split it did not beat the physiology score.

Implemented now: batch synthetic streams, offline update, scenario slider. Not implemented: a live wearable, a hospital EHR connection, or bidirectional therapy control.

Model

The Bergman minimal model is the twin. G is glucose (mg/dL), X is insulin action (1/min), SI is insulin sensitivity (1/min per µU/mL), SG is glucose effectiveness (1/min), and Ra is meal glucose appearance (mg/dL/min).

dG/dt = –(SG + X) G + SG Gb + Ra(t)

dX/dt = –p2 X + p2 SI (I – Ib)

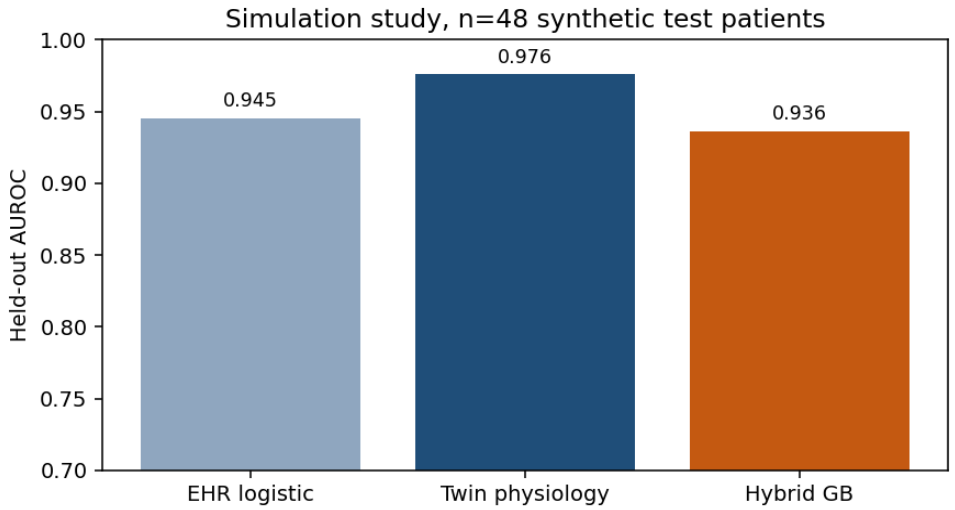
The generator adds a sleep and heart-rate stress term and sensor noise that the twin does not contain, so the twin is misspecified relative to the data. All 160 patients are synthetic. Parameter ranges are literature-informed bounds, not samples from ICMR microdata.

Measured results — simulation study

Train 112 / test 48 synthetic patients, seed 2026. Spike prevalence in the test split was 0.73, so accuracy alone would be misleading. Bootstrap intervals are percentile intervals from 1,000 resamples of the same 48 patients.

Model	AUROC	95% bootstrap	AUPRC	Brier	F1
EHR logistic	0.945	0.870–0.998	0.980	0.115	0.892
Twin physiology	0.976	0.926–1.000	0.992	0.068	0.925
Hybrid gradient boosting	0.936	0.853–0.994	0.978	0.089	0.930

Peak forecast error on the test split: MAE 17.1 mg/dL, RMSE 25.7 mg/dL. The intervals are wide because the test set is small. These are not clinical performance figures.



What a judge can run

- Public repository: Parth_Pariwandh_Jadavpur_University.
- python scripts/run_experiment.py regenerates the cohort, twin table, and metrics.
- streamlit run app.py opens the clinician scenario view.
- docs/clinician_dashboard.html is a static view of two held-out virtual patients.
- docs/architecture_diagram.pdf and docs/project_presentation.pdf are the required documents.

Licence: MIT. No patient identifiers, credentials, or private records are in the repository.

Limitation to say aloud: the video link is a submission item the author must record and upload. This prototype does not claim a completed clinical study.